

MOHAMED MUFASSIR

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PROFESSIONAL SUMMARY

3rd-year Computer Systems Engineering undergraduate (SLIIT) specializing in embedded systems, robotics, real-time control systems and Network Engineering. Proven ability to design and build intelligent hardware-software systems — from AVR-programmed autonomous robots to IoT-connected vehicles with live video streaming. Supplemented by industry internship experience in software development. Passionate about robotics, industrial electronics, and building embedded solutions that operate reliably in the real world.

EDUCATION

BSc (Hons) in Computer Systems Engineering

Nov 2023 – Expected Nov 2027

Sri Lanka Institute of Information Technology (SLIIT) | GPA: ~3.0

Relevant Modules: Real-Time Systems, Digital Logic Design, Control Theory, Computer Architecture, Data Structures & Algorithms, Computer Networks

Advanced Level Examination – Physical Science Stream

2022 (2023)

Combined Mathematics – B, Physics – C, Chemistry – C

EXPERIENCE

Software Engineering Intern — ZILLIT (Startup), Nittambuwa

Nov 2025 – Feb 2026

- Led sole development of a cross-platform Point of Sale (POS) desktop application using Electron.js and TypeScript, designed as part of a broader IoT ecosystem (desktop + mobile integration)
- Built the front-end of an enterprise-level financial management system using Vue.js, integrating with a Java Spring Boot backend
- Worked with React, TypeScript, Vue.js, Electron.js, PostgreSQL, and Docker in a fast-paced startup environment
- Collaborated on system architecture decisions for scalable, production-ready software

Freelance Web Developer — Self-Employed, Local & Online

2024 – Present

- Designed and developed business websites for local clients including a jewelry shop, hardware store, restaurant, and an indoor playground booking platform
- Delivered end-to-end solutions covering UI design, branding, and front-end development
- Technologies used: React, HTML/CSS, JavaScript

TECHNICAL SKILLS

- Languages:** C/C++, Embedded C, AVR Assembly, Python, JavaScript, TypeScript, Rust(Basic)
- Frontend:** React, Vue.js, Electron.js, HTML/CSS/JS
- Backend & DB:** Spring Boot (Java), Node.js, Express.js, PostgreSQL, My SQL, Docker (basic)
- Embedded & Hardware:** AVR (AT mega), Arduino, ESP32, Raspberry Pi, UART / SPI / I2C protocols, Oscilloscope & lab instruments, Digital Logic Design, Sensors & Actuators, PCB Design, 3D CAD Design
- Tools:** Git, Linux OS, Arduino IDE, Proteus, MATLAB, Fusion 360, VS Code, Easy EDA

PROJECTS

PID-Based DC Motor Speed Control System (Raspberry Pi + Python)

2026

- Designed a real-time closed-loop control system using PID algorithm with encoder feedback for precise RPM measurement and regulation
- Extended the project with a Python desktop GUI application to visualize live motor behavior, RPM graphs, and PID tuning parameters in real time
- Achieved stable, accurate motor control — successfully demonstrated and documented system performance
- Stack: Python, Raspberry Pi, PID control, encoder interfacing, data visualization

AVR Autonomous Line-Following Car with Smart Parking (Arduino Uno)

2026

- Programmed an autonomous line-following robot using AVR (AT mega) in Microchip Studio with 3 IR sensors for line tracking and crossline detection logic
- Implemented a parking bay detection system using an IR receiver — upon single crossline detection, the car enters parking search mode, detects the bay, and parks autonomously
- On double crossline detection, the car exits parking mode and resumes normal line following, demonstrating a full FSM-based behavioral control system
- Stack: AVR (AT mega), Arduino Uno, Microchip Studio, Embedded C, IR sensors

ESP32-CAM Remote-Controlled Car with Live Video Streaming

2025

- Built a WiFi-controlled RC car with live video streaming using the ESP32-CAM module as both the access point and video server
- Developed a mobile app and a web-based controller (live on Vercel) allowing real-time driving and camera feed viewing from any device on the network
- Integrated live MJPEG video stream with directional controls, synchronizing car movement and video feed over ESP32's internal WiFi
- Stack: ESP32-CAM, Embedded C, JavaScript, Vercel, WiFi/HTTP streaming

IoT Smart Surveillance & Access Control System (ESP32)

2024

- Built a smart surveillance system integrating camera module with Telegram bot for real-time intrusion detection and remote door lock/unlock via commands
- Implemented UART/I2C communication protocols and real-time IoT messaging for reliable remote control
- Stack: ESP32, C/Embedded C, Telegram Bot API, IoT communication

FSM-Based Sequence Detector

2024

- Designed a pseudo-random sequence generator using XOR feedback logic and PISO shift register serialization
- Implemented a Finite State Machine (FSM) for pattern detection with synchronized clocking

Learning Management System (Spring Boot)

2024

- Developed RESTful backend APIs for authentication and course management with scalable database integration

CERTIFICATIONS & LEARNING

- **Introduction to Embedded Systems Software and Development Environments** – University of Colorado Boulder (Coursera) - embedded software development workflows, GCC toolchain, Linux development environments, version control using Git, and ARM-based embedded system concepts
- **The Complete Full-Stack Web Developer Bootcamp** — Dr. Angela Yu, Udemy (HTML, CSS, JS, Node.js, React, PostgreSQL)

INTEREST AREAS

Embedded Systems | IoT | Robotics & Control Systems | Network Engineering | Telecommunications | Industrial Electronics | Software Engineering

REFERENCES

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